

Econ 139: Labor Economics

Trade (Theory)

David Arnold



Trade

- The Clark Center at University of Chicago regularly surveys many of the leading economists in the world on a variety of topics. In 2012, they asked the following question:

Freer trade improves productive efficiency and offers consumers better choices, and in the long run these gains are much larger than any effects on employment

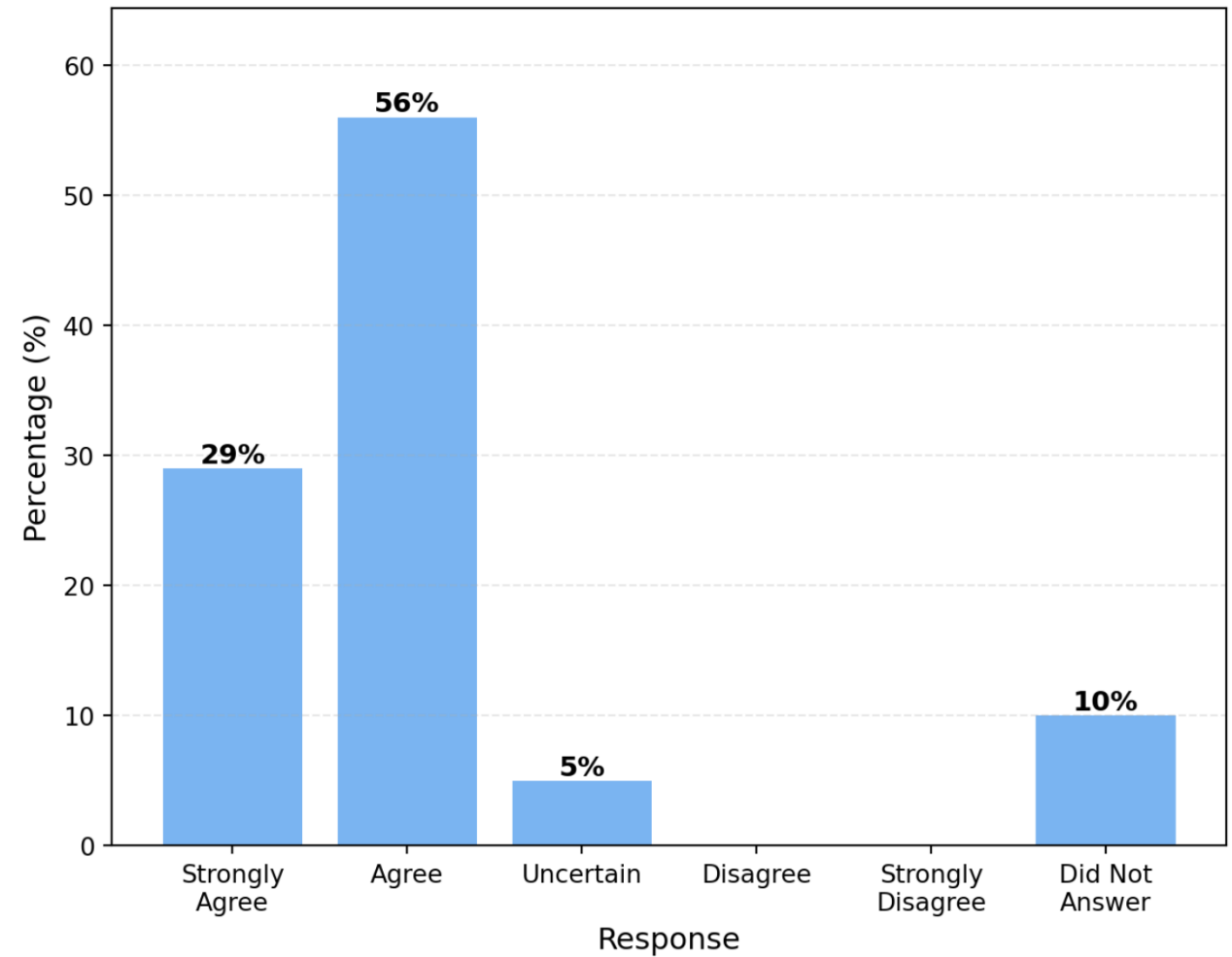


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Trade

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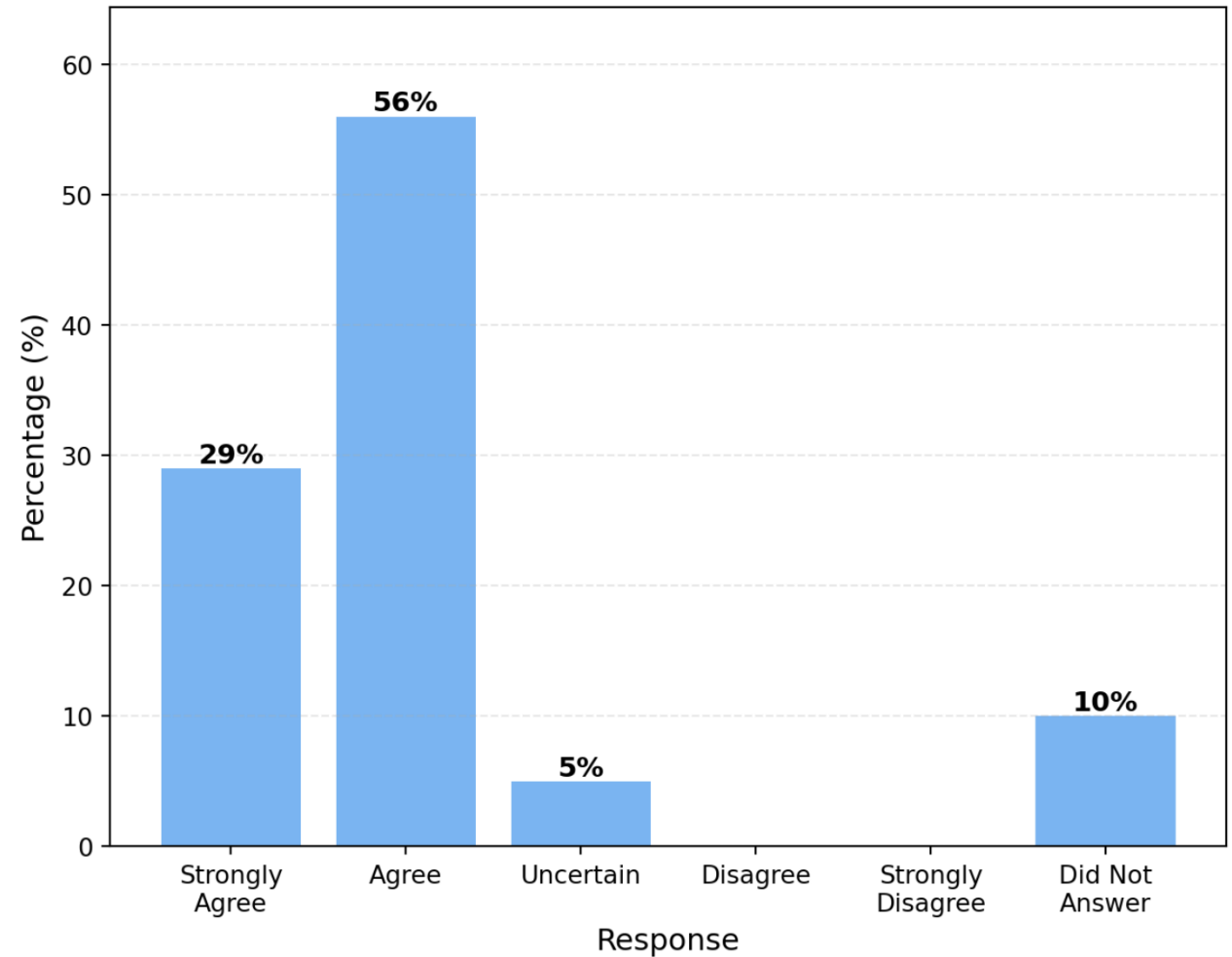
Survey Responses on Free Trade from Economists



Trade

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- These experts were also asked about the North America Free Trade Agreement (NAFTA) in particular and universally agreed it improved welfare for US citizens

Survey Responses on Free Trade from Economists



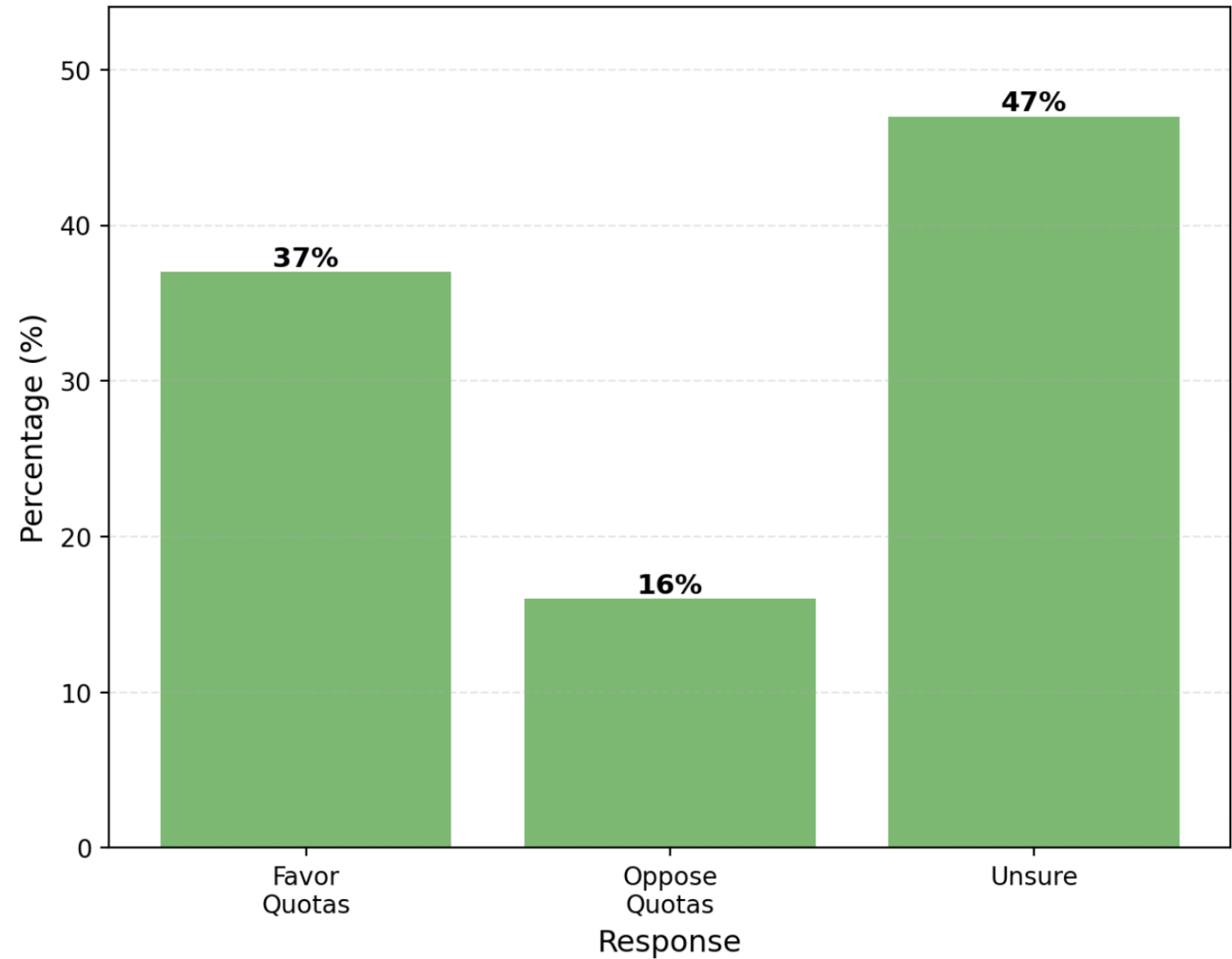
Public Opinion

- Public opinion on Trade is much more mixed. In 2012, when the American National Election Studies (ANES) asked
- *Some people have suggested placing new limits on foreign imports in order to protect American jobs. Others say that such limits would raise consumer prices and hurt American exports. Do you FAVOR or OPPOSE placing new limits on imports, or haven't you thought much about this?*

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Public Opinion on Import Quotas

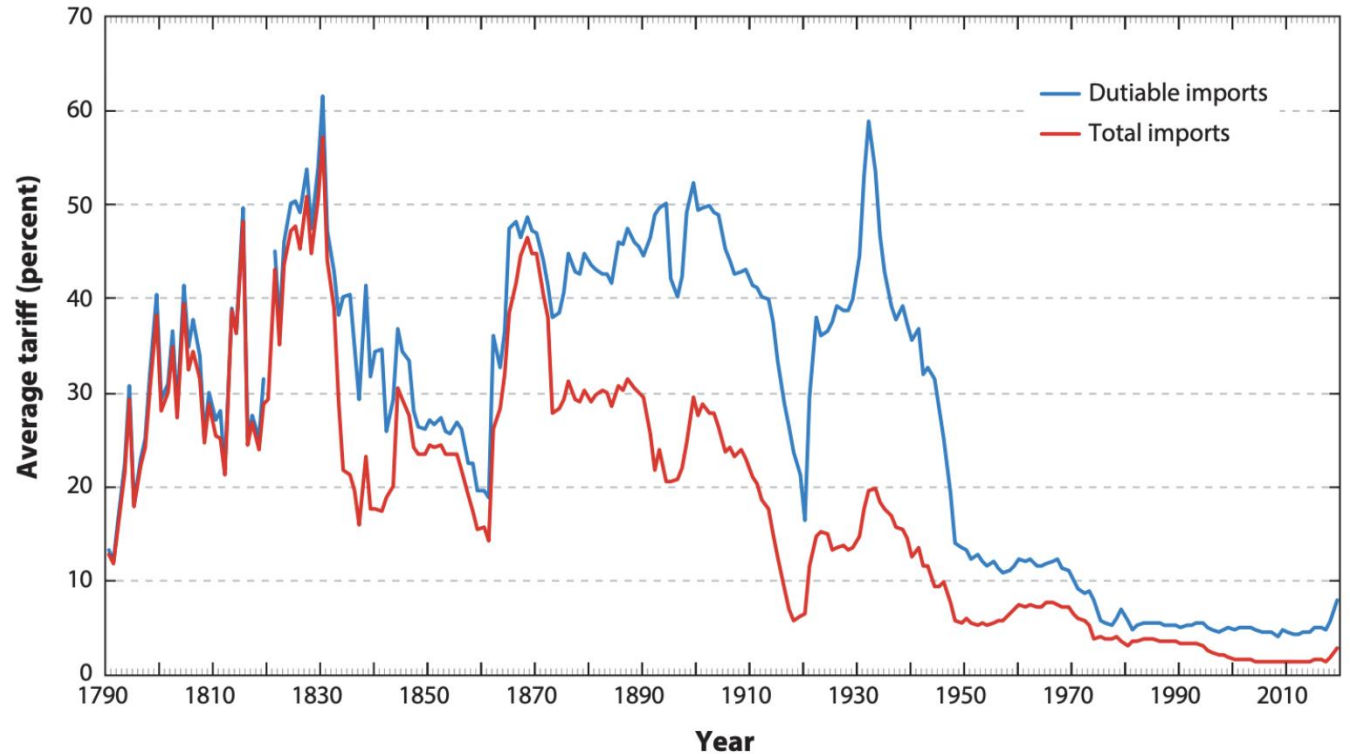


Trade Policy

- Uncertainty is (somewhat) reflected in policy as well
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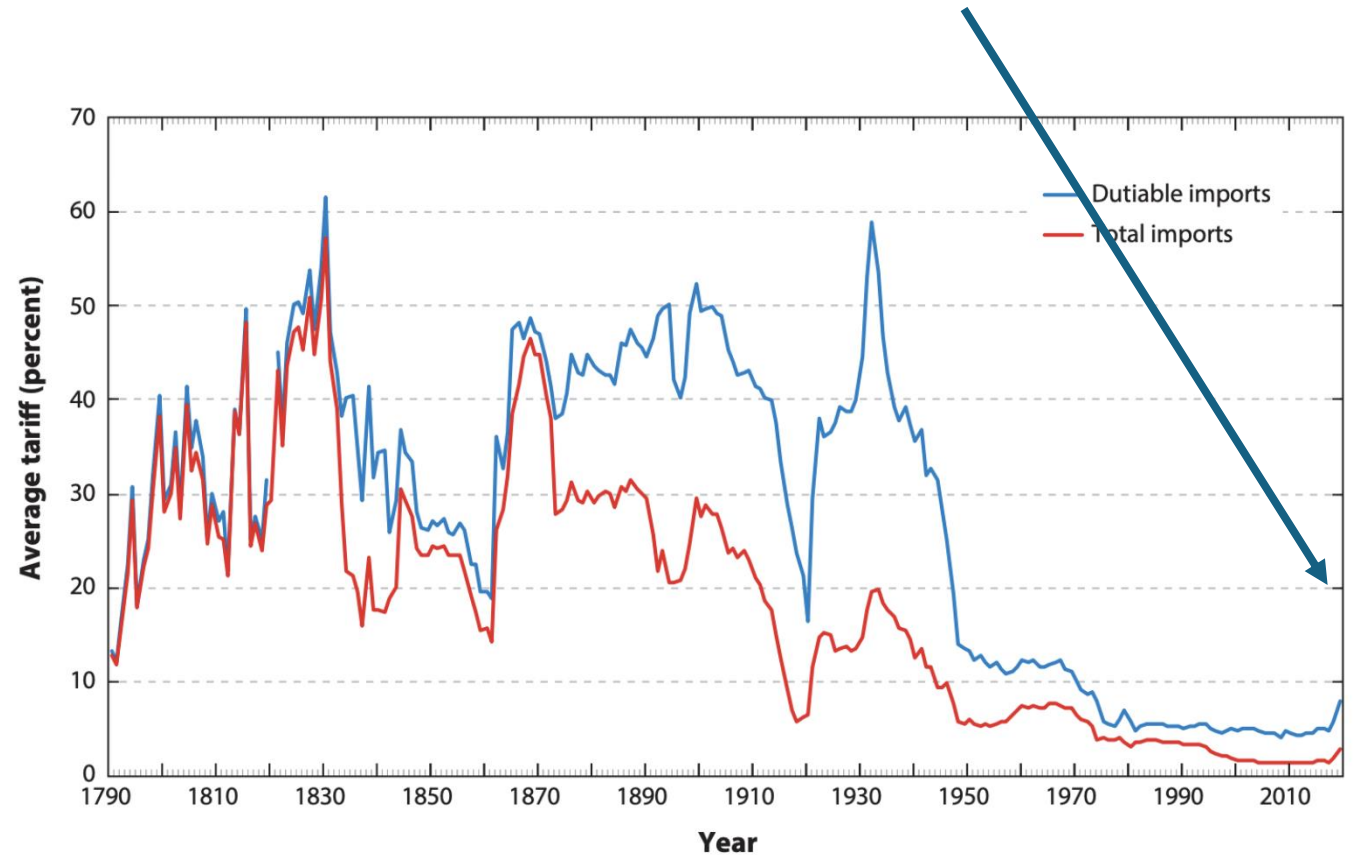


Source: Irwin (2020)

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Tariff rate under new Trump tariffs estimated to be just below 20 percent



Source: Irwin (2020)

Winners and Losers

- Something that has been recognized for a long time is that trade creates winners and losers.
- Ricardo (1815) discusses how the winners of the corn laws (which placed tariffs on corn) raised the price of corn to the benefit of landowners and the detriment of manufacturing workers

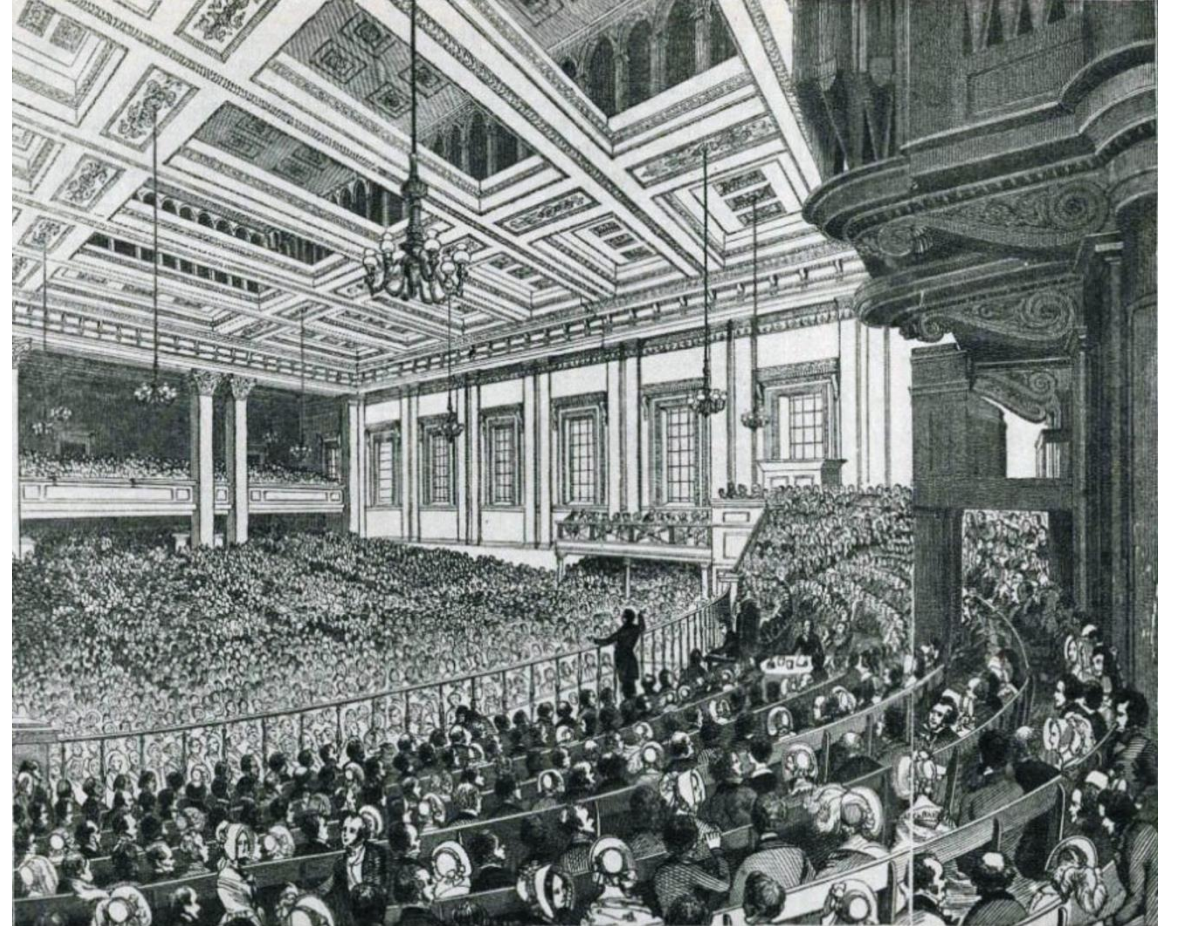
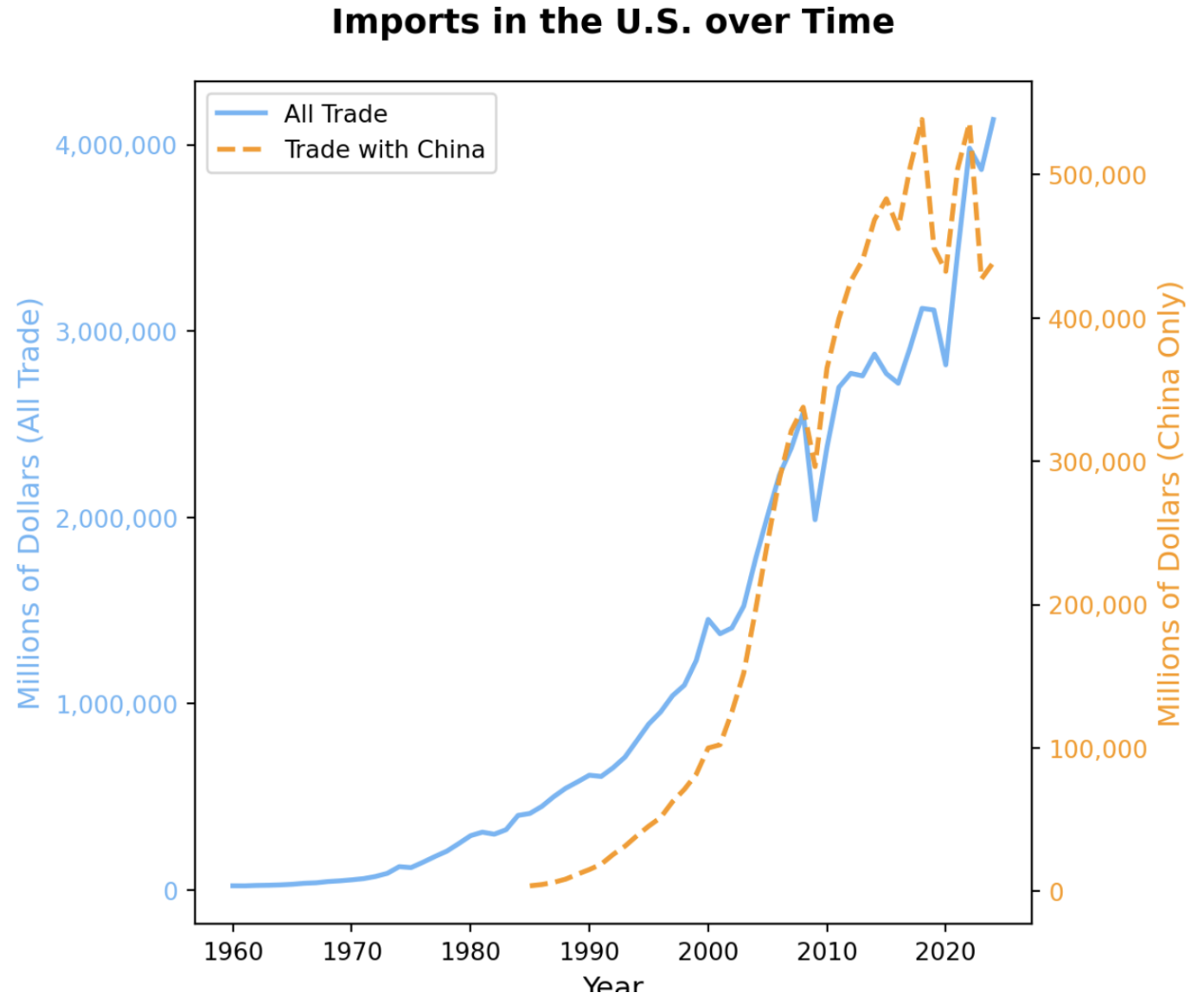


Image: A meeting of the anti-corn league

Trends in Trade

- The winners and losers debate received renewed attention with the rise of trade with China
- In 2001, China joined the World Trade Organization (WTO) which led to steep increases with trade from China



Comparative Advantage

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Name one proposition in all of the social sciences which is both true and non-trivial.

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- Paul Samuelson's answer was the concept of comparative advantage
 - **True:** we will see mathematically this is true
 - **Non-trivial:** Samuelson: "That [comparative advantage] is not trivial is attested by the thousands of important and intelligent men who have never been able to grasp the doctrine for themselves or to believe it after it was explained to them"

Comparative Advantage

- Comparative Advantage
 - Even though one country may be more productive (in everything) relative to another, the two can still gain from trade
- It's intuitive that if Country A can produce clothing more cheaply than Country B, and Country B can grow food more cheaply than Country A, then both countries could benefit from trade.
- But what remains true is that they will benefit from trade even if one country is better at producing both goods. This is the power of the principle of comparative advantage.

Comparative Advantage

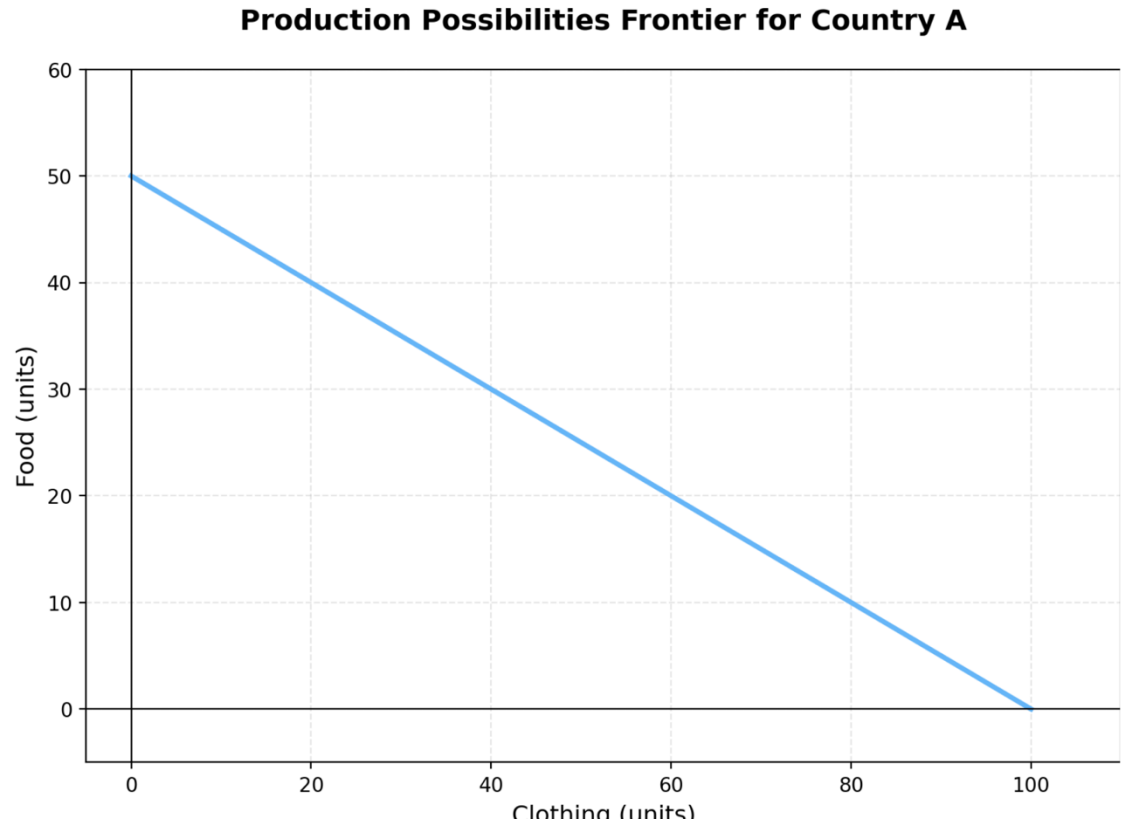
- Let's put some concrete numbers on this
- We will have two countries
 - Country A will be able to produce less of both clothing and food
 - Country B will be able to produce more of both clothing and food
- Despite this, we will find that trade benefits both Country A and Country B

The Production Possibilities Frontier (PPF)

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- Let's assume Country A's PPF is shown here

The Production Possibilities Frontier (PPF)

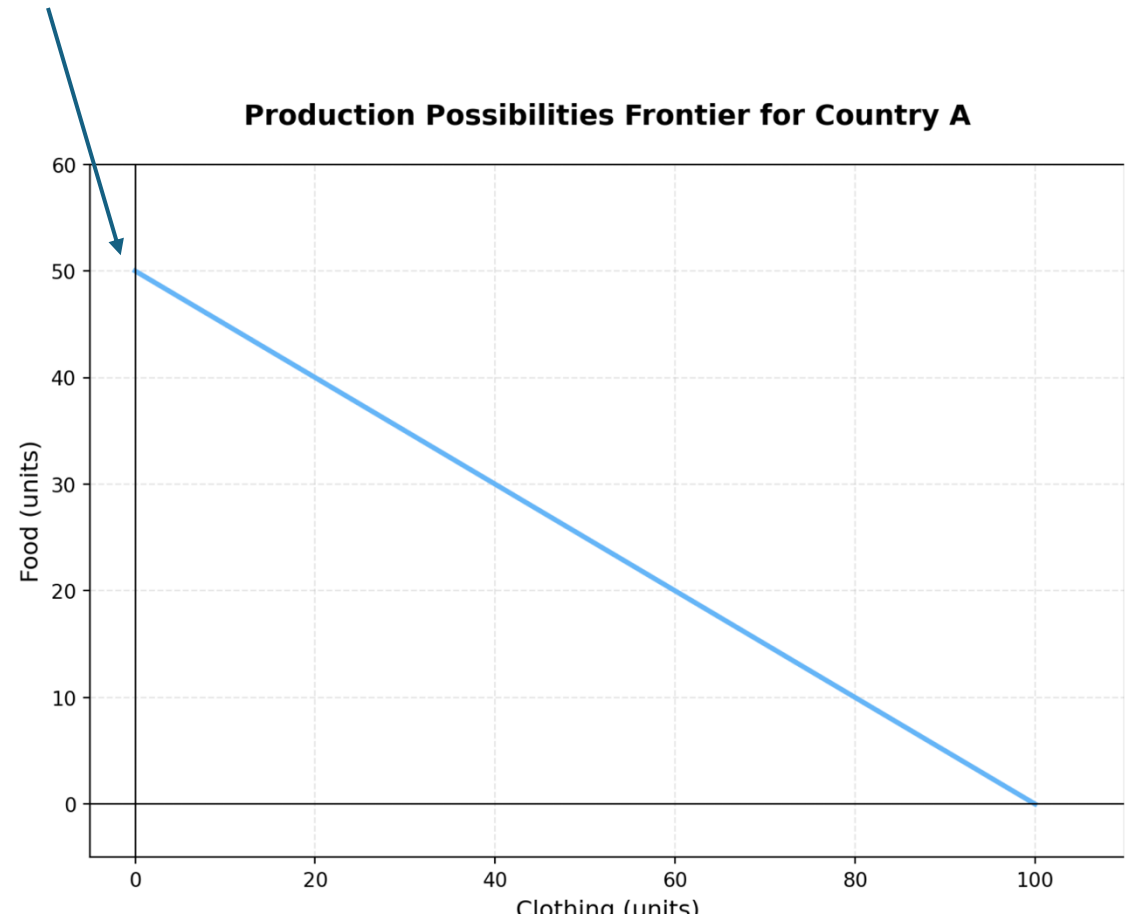
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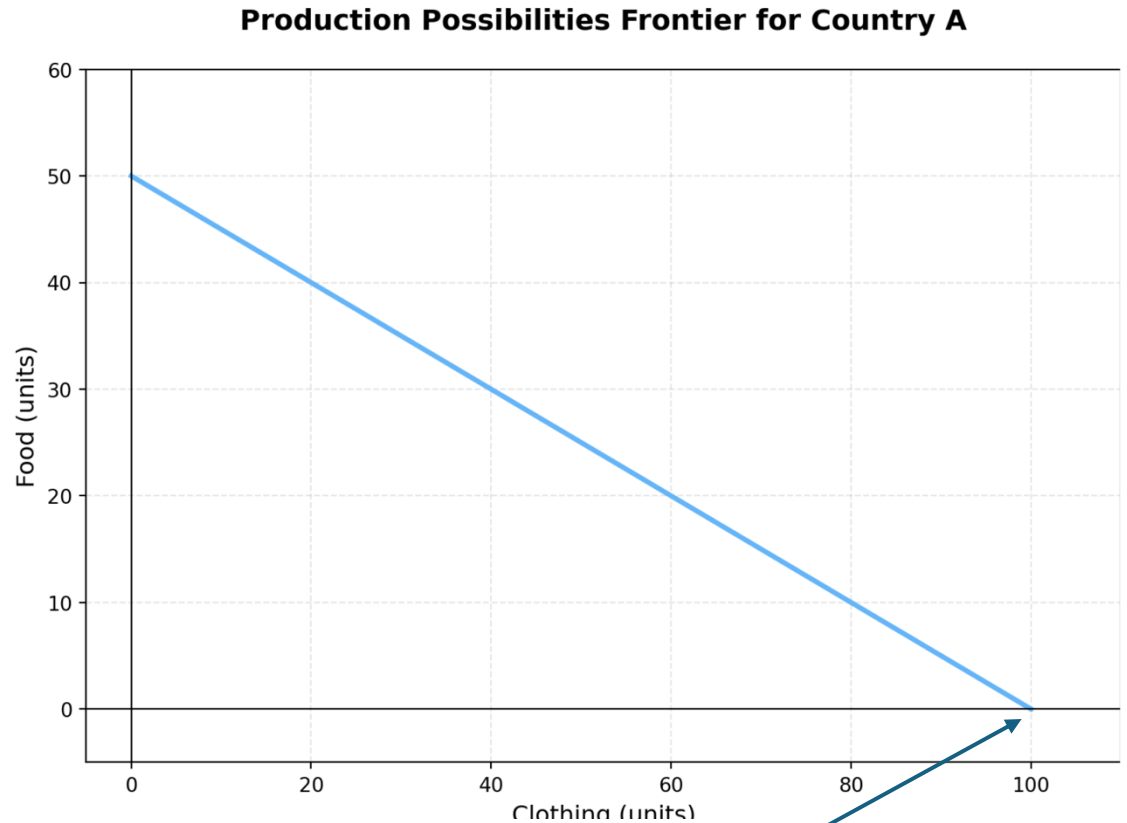
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Amount of Food if all resources put into foods



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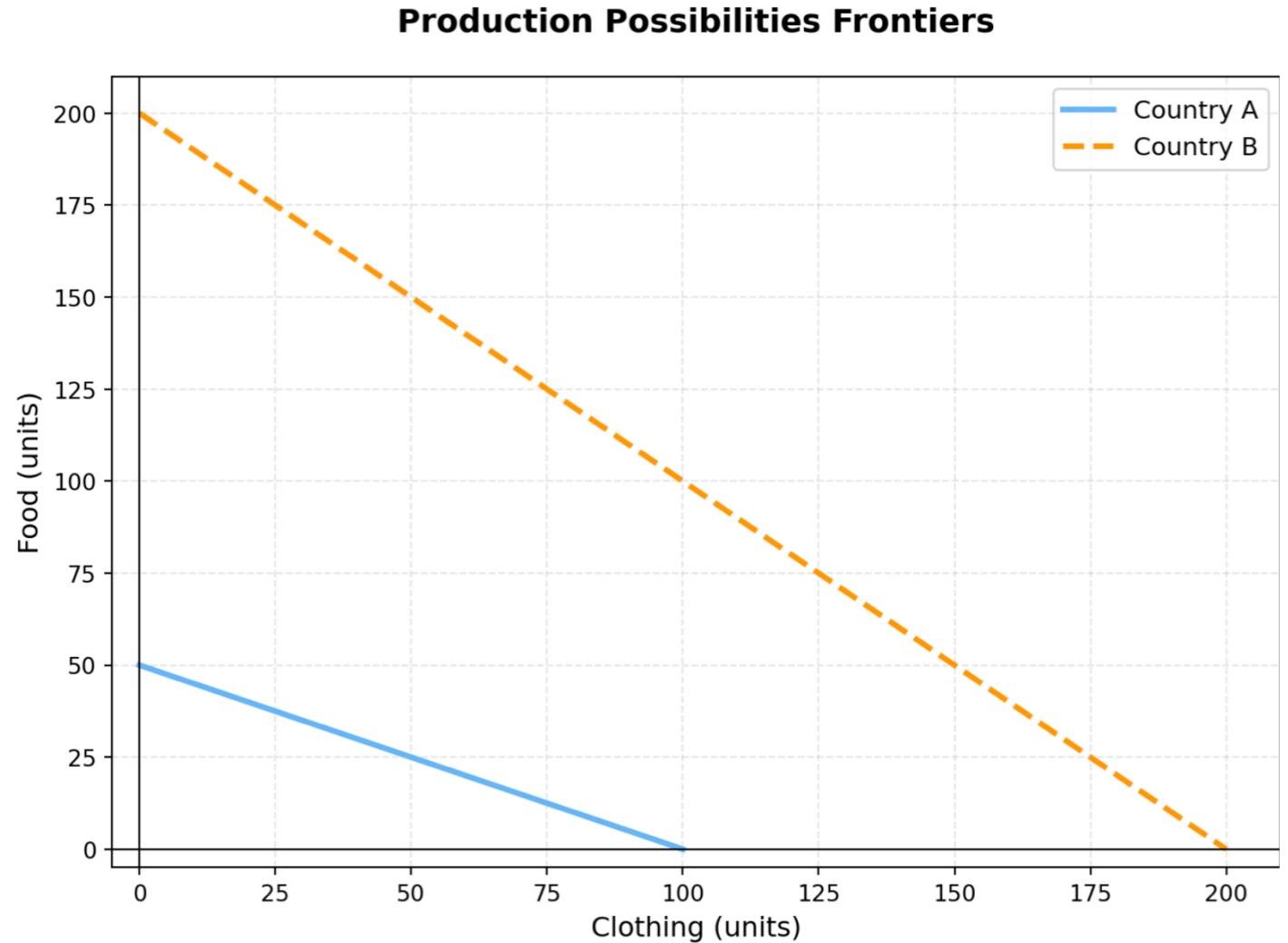
Amount of Clothing if all resources put into foods

The Production Possibilities Frontier (PPF)

- Now let's add Country B, which can produce more than Country A

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Comparative Advantage – Opportunity Costs

- The key to understanding comparative advantage is to understand that when each country produces a good, it faces an opportunity cost
 - If country A wants one more unit of clothing, it must give up 0.5 units of food
 - If country B wants one more unit of clothing, it must give up 1 unit of food

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 - If country A wants one more unit of clothing, it must give up 0.5 units of food
 - If country B wants one more unit of clothing, it must give up 1 unit of food
- Country A has a **comparative advantage** in clothing, because it must give up fewer units of food for an additional unit of clothing
 - The flip side is also true: country B has a comparative advantage in food because it must give up less clothing for an additional unit of food

Comparative Advantage – Opportunity Costs

- Where a country chooses to be on its PPF depends on preferences
- What we will show is that with trade, the country can consumer **strictly more of both goods**
 - This is true for both the less-productive country (Country A) and the more productive country (Country B)

PPF with Trade

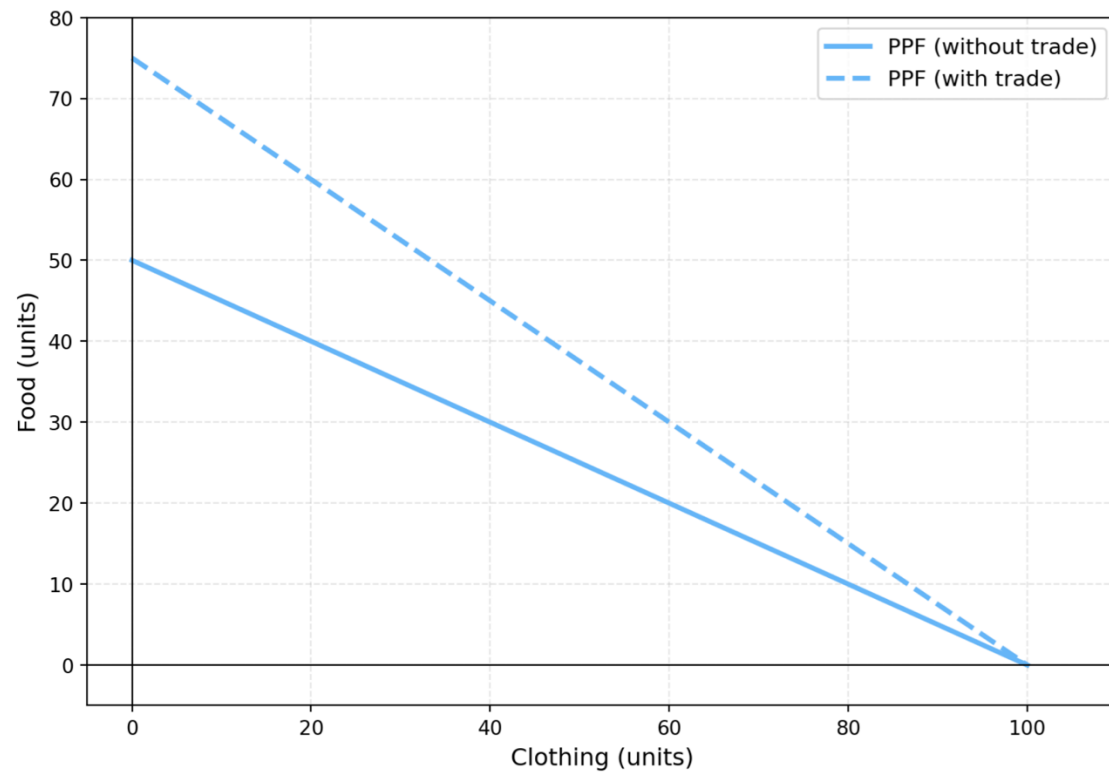
- Let's assume the price of exchange is 1 unit of clothing for 0.75 units of food
 - Country A will want to trade clothing. Before, in autarky (i.e. no trade), giving up 1 unit of food would equal 0.5 units of food. Trade is a better deal!

PPF with Trade

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 - Country B will want to trade food. Before, in autarky (i.e. no trade), giving up 1 unit of food would equal 1 unit of clothing. Now they only have to give up 0.75 units of food for 1 unit of clothing
- We can visualize the gains from trade by considering the new PPF assuming these prices

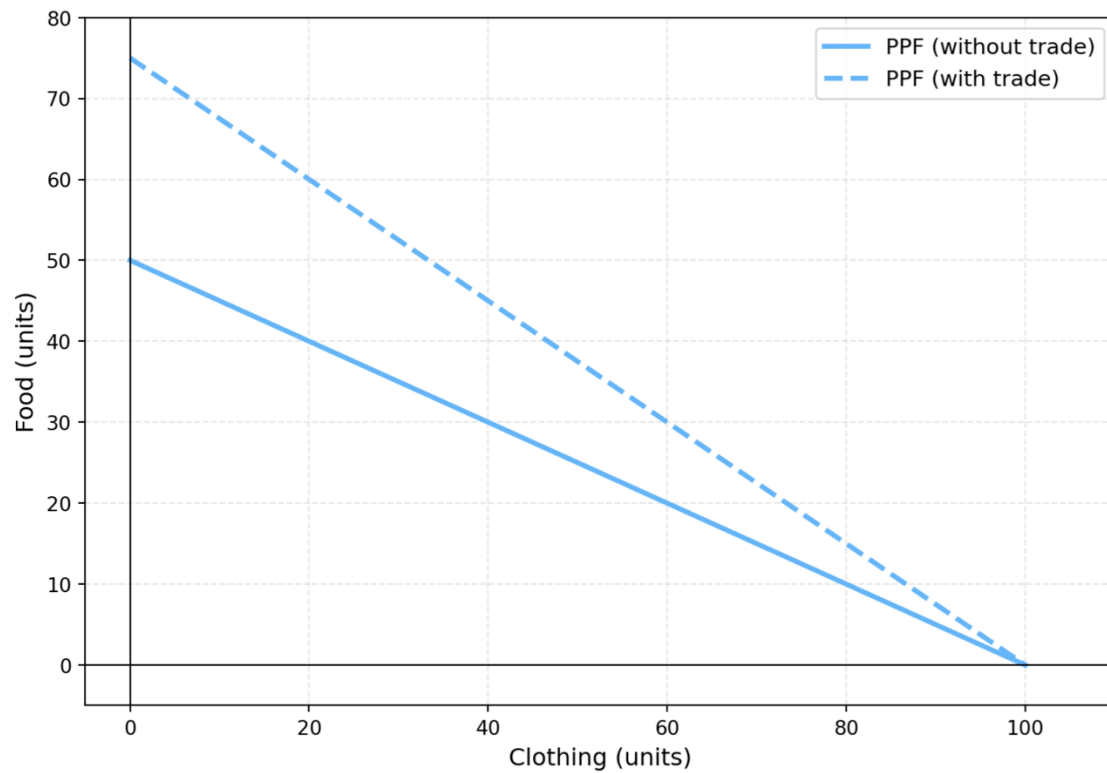
PPFs with Trade

Country A: Production vs Consumption Possibilities

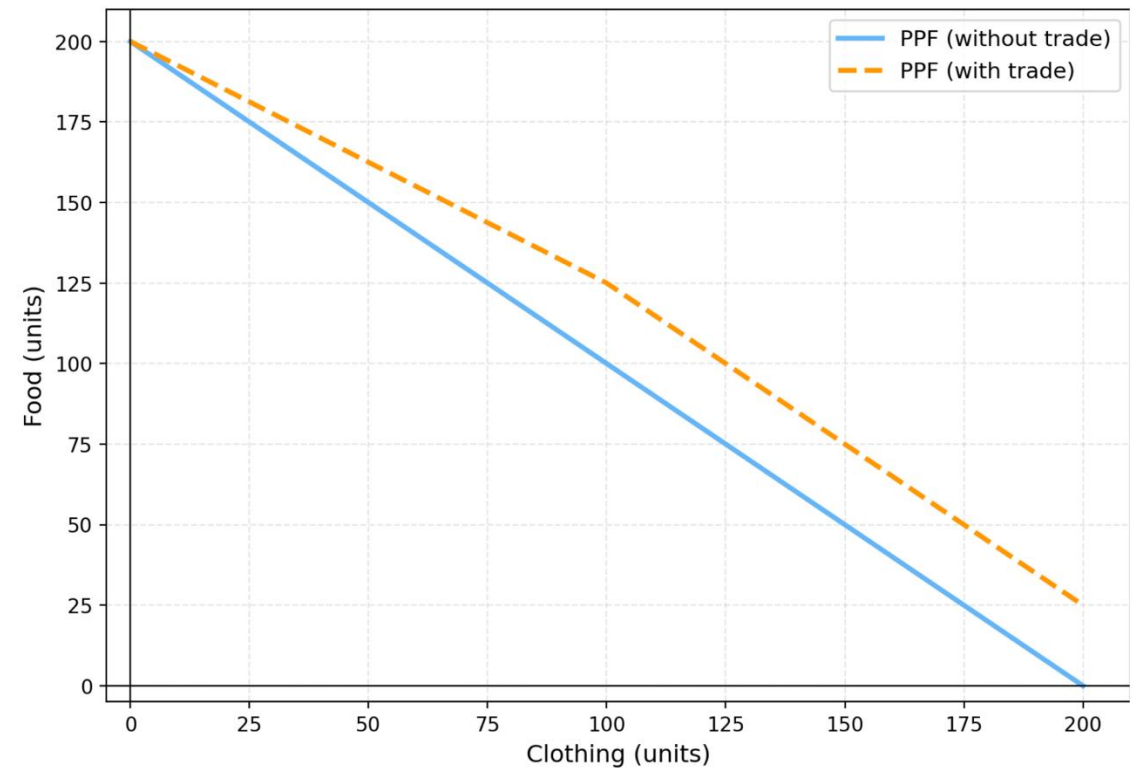


PPFs with Trade

Country A: Production vs Consumption Possibilities



Country B: Production vs Consumption Possibilities



Comparative Advantage

- This is a pretty simple model, but a powerful one
 - The concept of comparative advantage has shaped support for free trade for centuries
- However, it has long been recognized that trade will create distributional effects (i.e. winners and losers)
 - In Country A: workers in food industry will lose jobs
 - But overall economy does expand, so these “losers” to trade could be compensated theoretically
- This is the primary tension in debates about trade: how do we tradeoff the aggregate gains against the concentrated losses for some workers

Further Reading

- In this class, as our focus is on labor, we will consider these two primary mechanisms
 - Aggregate growth due to trade: we will show this is surprisingly difficult to identify, but has good theoretical backing
 - Distributional effects: we will find certain places in US have suffered long-term negative impacts due to trade
- This simple model of trade cannot explain many empirical facts about trade in the modern era
 - Why is there so much intra-industry trade? (Paul Krugman won the Nobel Prize in 2008 for his work on this question)



Dec 1 — Write down your first and last name